Vapor Stone
RAIL SYSTEMS
A Wabtec company

10 655 Henri-Bourassa Blvd., West
Saint-Laurent (Québec) Canada
H4S 1A1

Tel: 514.335.4200
Fax: 514.335.4231

www.wabtec.com

Vapor Stone Rail Systems

Generic Software Documentation Project

Project # 7260

Boot Loader Application

Software Integration Test Plan (SITP)

Contract Number	TBD	CDRL Number	TBD
		Customer Ref. Number	TBD

NOTICE
Plans, specifications and engineering data submitted herewith or which may be submitted, remain the property of **Vapor Stone Rail Systems** and shall not be reproduced or altered or used in whole or in part except by express consent of **VSRS**.

AVIS
Les plans, spécifications et données techniques soumis ci-joints, ou qui peuvent être soumis, restent la propriété de **Vapor Stone Rail Systems** et ne doivent pas être reproduits, altérés ou utilisés, totalement ou partiellement, sans le consentement express de **VSRS**.

	Name / Title	**Signature / Date**
Prepared :	José Trejo Software Developer	Jose Emanuel Trejo de la Torre 2011-09-22
Verified :	Éric Malo, Eng. System Engineer	2011-09-23
Approved :	Martin Ouellet, Eng. Software Development Supervisor	2011-09-23

Document No.	**201111580**	Revision	**1**
		Date (yyyy-mm-dd)	**2011-09-22**

Revision List

Rev.	ECD#	Date	By	Description of changes
1	--	2011-09-22	José Trejo	First Release.
0.2	--	2011-09-21	José Trejo	Implemented comments for internal inspection I-1804 2nd round
0.1	---	2011-09-19	José Trejo	Implemented comments from internal inspection I-1804 interaction 1.
0	---	2011-09-02	José Trejo	Original Issue

Preface

This Software Integration Test Plan (SITP) is prepared in accordance with IEEE-829-1998, with slight adaptation to the format. It has been adapted to combine the 3 documents: Test Plan, Test Design Specification, and the Test Procedure Specification.

Table of Contents

REVISION LIST **2**

PREFACE **3**

TABLE OF CONTENTS **4**

LIST OF FIGURES **6**

LIST OF TABLES **6**

1 TEST PLAN **7**

1.1 Test Plan Identifier 7
1.2 Introduction 7
1.3 Objectives 7
1.4 Scope 7
1.5 Definitions, Acronyms and References 7
1.5.1 Definitions 7
1.5.2 Acronyms 8
1.5.3 References 8
1.6 Test Items 9
1.7 Features to be tested 9
1.8 Features not to be tested 9
1.9 Approach 9
1.10 Item Pass/Fail Criteria 10
1.11 Suspension Criteria and Resumption Requirements 10
1.11.1 Suspension Criteria 10
1.11.2 Resumption Requirements 10
1.12 Test Deliverables 10
1.13 Testing tasks 10
1.14 Environmental Needs 10
1.14.1 Hardware 10
1.14.2 Software 10
1.15 Responsibilities 10
1.16 Staffing and training needs 10
1.17 Schedule 10
1.18 Risks and contingencies 11
1.19 Approvals 11

2 TEST PROCEDURE SPECIFICATION **12**

2.1 Purpose 12
2.2 Special Requirements 12
2.3 Procedure Steps 13
2.3.1 Test Log 13
2.3.2 Test Setup 13
2.3.3 PTA Installation 14
2.3.4 PTA Simulator Installation 14
2.3.5 Boot Loader Programming 14
2.3.6 DMS Application Corruption 15
2.3.7 Emulator Setup 16
2.3.8 Other Information about Test 16

3 PROCEDURE **17**

3.1 Login/Logout 18
3.2 Portable Test Application (PTA) Communication Error 21

3.3 Fault LED & Self Test 21
3.4 Target Module Application upgrade 23
3.5 Software Version 27
3.6 Static RAM Memory Dump 28
3.7 Request DMS Code Status 29
3.8 Door Control Access 30
3.9 DMS Boot Loader upgrade 31
3.10 Verification of the Connection State 32
3.11 Controller ID 32
3.12 Static Numerical Requirements 34

APPENDIX A: TEST REPORT SHEET 37

List of Figures

Figure 2-1: External Interface Diagram 12
Figure 2-2: Test Setup 13

List of Tables

Table 2-1: Hardware Required 12
Table 2-2: Software Required 13
Table 3-1: Memory Mapping Example 35

1 TEST PLAN

1.1 Test Plan Identifier

201111580 Rev. 1

1.2 Introduction

The Software Integration Test Plan (SITP) will be created for each of the SCIs and includes 3 major sections:

- **Test Planning**

  Identifies the tests required to verify and validate each SCI. Its content shall be based and comply with the requirements of IEEE STD 829, standard for Software Test Documentation.

- **Test Design Specification**

  Describes the design of the tests to be performed in order to validate the requirements of each SCI. This is merged with the test planning and the Test Procedure Specification.

- **Test Procedures Specification**

  Describes the procedures that need to be executed in order to validate the requirements of each SCI.

1.3 Objectives

The objective is to plan the way the Software Requirements Specification (SRS) will be tested as well as to define the tests required to validate Door Module System (DMS) Boot Loader.

1.4 Scope

The scope of this document is to cover the requirements of the DMS Boot Loader to verify the DMS hardware and allow the DMS application to be upgraded.

1.5 Definitions, Acronyms and References

1.5.1 Definitions

Boot Loader :	Software starting the target application of a door module.
Communication Gateway Module:	A specific processor based electronic module running a Communication Gateway Module application. It acts as gateway between the Door Control Module application and the Car Network/Portable Test Application.
Door Control Unit:	The final unit that is used to control one or more door panels. This unit is normally made of one or two door control modules running the Door Control Module Application and possibly one Communication Gateway Module Application.
Door Control Module:	A specific processor based electronic module running a door Control Module Application. It is used for controlling the Door motor.

1.5.2 Acronyms

CPU	Central Process Unit
CRC	Cyclic Redundancy Check
DCM	Door Control Module
DCU	Door Control Unit
DMS	Door Module System
HW_COM	Hardware Communication
I/O	Input/Output
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers
LED	Light-Emitting Diode
nVolatileMemory	Non-Volatile Memory
PC	Personal Computer
PTA	Portable Test Application
RAM	Random Access Memory
RX	Receive
SACS	Software Applications Configuration Specifications
SCI	Software Configuration Item
SDD	Software Design Description
SPR	Software Problem Report
SQAP	Software Quality Assurance Plan
SQAP	Software Quality Assurance Plan
SRS	Software Requirements Specifications
SRTM	Software Requirements Traceability Matrix
STP	Specific to Project
SVVP	Software Verification and Validation Plan
TBD	To be defined
TS	Technical Specifications
TX	Transmit
USB	Universal Serial Bus
VSRS	Vapor Stone Rail Systems

1.5.3 References

This section lists the number, title, revision, date, and source of all documents referenced in this Integration Test Procedure Specification.

Standards

IEEE Std 829-1998	IEEE Standard for Software Test Documentation, September 1998, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, February 2nd 2005, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

201111271	Software Requirement Specification, Generic Software Documentation Project – Boot Loader Application, Rev. 0.1.
201111456	Software Design Description, Door Monitoring and Control Unit –Boot Loader Application, Rev.0.1.
S17-10	Software Integration Test Procedure Template, Rev.2, 2002-01-15
SACS	Software Applications Configurations Specifications, Project's Specific.
SQAP	Software Quality Assurance Plan, Project's Specific.

SVVP	Software Verification and Validation Plan, Project's Specific.

Customer's Document

TBD.

1.6 Test Items

All the test that make up the DMS Boot Loader will be tested during the test phase. The version tested will be placed under configuration management.

The Software Requirement Specification (SRS) of the DMS Boot Loader will provide the basis for defining the correct operation.

1.7 Features to be tested

The features to be tested are shown in Section "3 Procedure".

1.8 Features not to be tested

All the features of the DMS Boot Loader Application are tested.

1.9 Approach

The test personnel will use the system documentation, SRS and Software Design Description (SDD) to prepare all test design, case, and procedures specifications. This approach will verify the accuracy and comprehensiveness of the information in the documentation in those areas covered by the tests.

The test personnel will run the test procedures and complete the test reports. The test data are to be under configuration management.

The following software programs will be used in the test procedure and if they don't exist, they will need to be developed prior to executing the test procedure.

- Portable Test Application (PTA)
- Portable Test Application Simulator (PTA Simulator)
- DMS Application

The PTA Simulator runs on a PC and communicates using the serial port. This application allows receiving commands and data. It also allows sending valid and invalid commands to the DMS Boot Loader. This will help testing error handling.

The following features of the DMS Boot Loader available for the PTA will be tested with the PTA Simulator.

- Valid Login/Logout
- Invalid Login/Physical disconnection
- Command error DMS Boot Loader to PTA
- Communication Test between PTA and DMS Boot Loader
- Display and select Controller ID

The following features of the DMS Application will be tested with the PTA.

- Request the code status of the DMS Application
- Download Static RAM Content
- Maximum of 50% CPU usage
- Upload a new version of the DMS Application
- Request Versions of the DMS

Performing a visual inspection of the DMS will test the following features of the DMS Boot Loader:

- DMS not able to Open or Close the Door

- Fault and Warning LED functionality
- DMS Boot Loader and DMS Application use less than 70% of installed memory (70% of nVolatile Memory, 70% of Volatile Memory, and 70% of Flash)

The Software Requirements Traceability Matrix (SRTM) will be used to trace all requirements from the SRS to the related test cases of the Integration Test Procedure Specification.

1.10 Item Pass/Fail Criteria

The Boot Loader software is to be considered acceptable if:

1. It meets all the criteria defined in the Software Requirements Specifications (SRS) – DMS Boot Loader or upon approval of non-conformities detected during the test by the customer.
2. After undergoing all the tests, the DMS Boot Loader software remains functional without corruption of settings

1.11 Suspension Criteria and Resumption Requirements

1.11.1 Suspension Criteria

The following criteria will cause a suspension of all testing activities:

- Failing one of the test cases.

1.11.2 Resumption Requirements

When a new version of the DMS Boot Loader software is transmitted to the test personnel after a suspension of testing has occurred, the procedure will be started again.

1.12 Test Deliverables

The following documents will be generated by the test personnel and will be placed under configuration management:

- DMS Boot Loader Test Plan
- DMS Boot Loader Software Test Report

1.13 Testing tasks

The testing tasks are defined in the SVVP[1] section 5 (V&V Processes)

1.14 Environmental Needs

1.14.1 Hardware

The materials and equipment used for this test plan are those called in the specific test procedures and test cases.

1.14.2 Software

All software applications used are those called in the specific test design and test procedure.

1.15 Responsibilities

The responsibilities are defined in the SVVP section 4.5.

1.16 Staffing and training needs

The staffing needs are defined in the SVVP section 4.6. No training is needed for this project.

1.17 Schedule

The schedule is defined in the SVVP section 4.2.

[1] Project Specific SVVP.

1.18 Risks and contingencies

The risks and contingencies are defined in the SQAP[2] section 14.

1.19 Approvals

Approvals' names and signatures are included in the cover page of this document.

[2] Project Specific SQAP.

2 TEST PROCEDURE SPECIFICATION

This section is the Software Integration Test Procedure (SITP) of the DMS Boot Loader for the "Generic Software Documentation Project" developed by Vapor Stone Rail Systems.

2.1 Purpose

The purpose of this Test Procedure Specification Section is to design and specify the tests required to validate the DMS Boot Loader Application that is part of the DMS.

The intended audience is the Vapor Stone Rail Systems software development and maintenance team members, and the customer.

Figure 2-1 shows the external interfaces of the DMS Boot Loader.

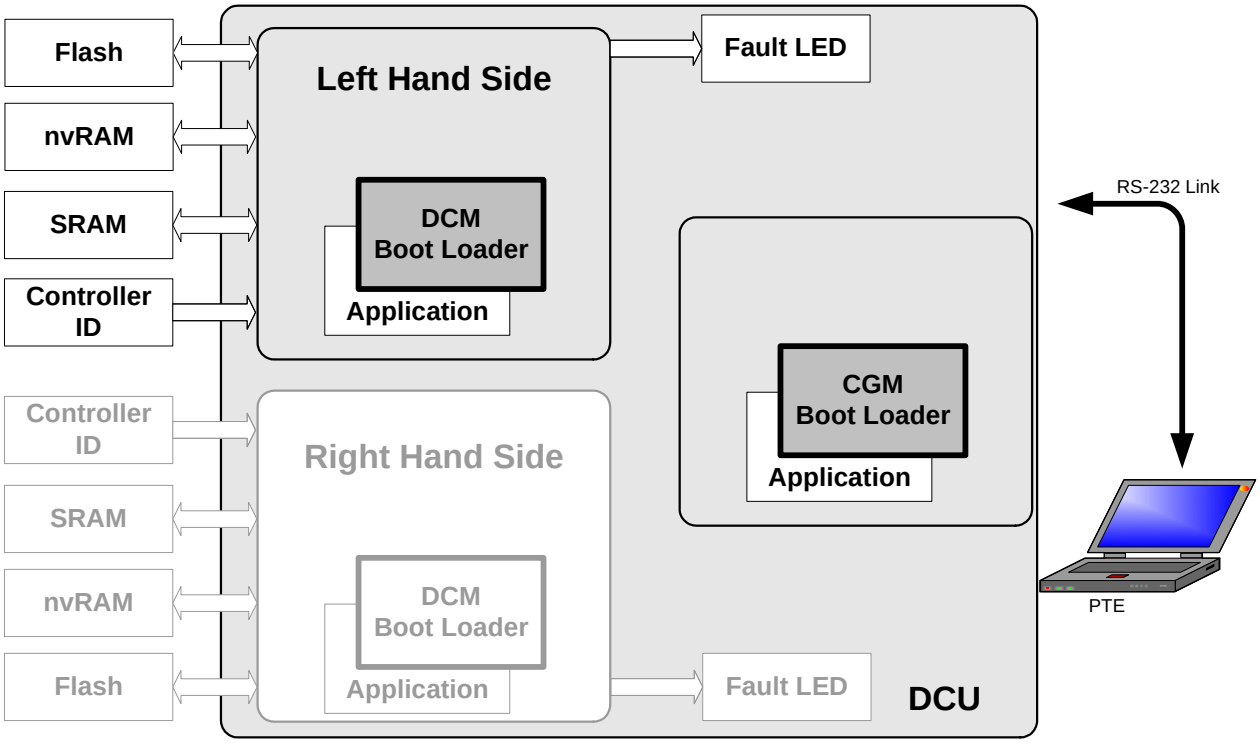

Figure 2-1: External Interface Diagram

2.2 Special Requirements

The following tables list the equipment, the software and the special files used in the testing:

Table 2-1: Hardware Required

Hardware	Version	P/N (specific to project)	Rev.
Computer, minimum PENTIUM, with Windows XP or Windows 7 (for PTA simulator application and PTA application)	--	--	--
Complete Door System: Specific to the project	--	--	--

Hardware	Version	P/N (specific to project)	Rev.
PTA cable DB9-PTA PLUG	--	--	--
Variable Power Supply (Voltage source min. 10A, 0-50V)	--	--	--
DMS Test Fixture	--	--	--
Banana cables	--	--	--
Multimeter	--	--	--
HiTex Development Tools DTrace 167/128k	--	--	--
Hitex DProbe-ST10	--	--	--
ST10CTRL + B76 CPU	--	--	--

Table 2-2: Software Required

Software	Version	P/N	Rev.
PTA Application	STP	STP	STP
Target Application	STP	STP	STP
Boot Loader Application	STP	STP	STP
PTA Simulator Application	STP	STP	STP
HiTOP Emulation Software	5.20.430	--	--

2.3 Procedure Steps

2.3.1 Test Log

A software integration test report will be generated to record the test results based on this document
The result of each test case log should be recorded in the last column, “P/F”,(see section 3 Procedure). Letter “P” or “F” will indicate Pass or Fail to the test.
A Test Report Sheet included in Appendix A of this document will be correctly filled (including Software Problem Report (SPR) if any), dated.
The tool used for (SPR) tracking is the ClearQuest database.

2.3.2 Test Setup

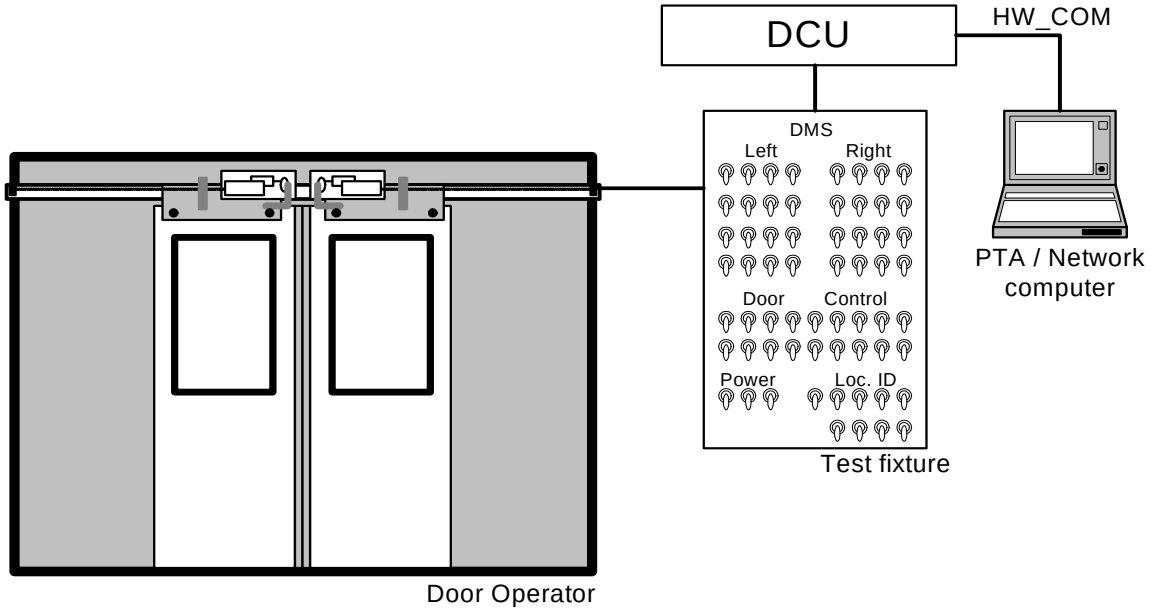

Figure 2-2: Test Setup

2.3.3 PTA Installation

The PTA software comes with an automated Windows-based installation program called SETUP. Don't try copying the files manually on your hard disk; instead please follow these steps to install the software:

1. Extract from the PTA installation zip file the setup.exe file that contains installation for the latest version of the PTA.
2. Run SETUP.EXE.
3. The setup program will then begin to copy and decompress files on the target drive. You will be prompted when it is necessary to insert subsequent diskettes.
4. The setup program will prompt you to terminate any application that might lock system resources or files. A good practice would be to stop any other application before running setup.
5. The next step will be to select a target directory for the program. It is recommended that the default location be used if possible.
6. Once the target directory has been selected and approved, the setup program initiates the installation procedure. Once the process is completed, the program will inform you if the installation was successful or not.

2.3.4 PTA Simulator Installation

The PTA Simulator software comes with an automated Windows-based installation program called SETUP. Don't try copying the files manually on your hard disk; instead please follow these steps to install the software:

1. Extract from the PTA Simulator installation zip file the setup.exe file that contains installation for the latest version of the PTA Simulator.
2. Run SETUP.EXE.
3. The setup program will then begin to copy and decompress files on the target drive. You will be prompted when it is necessary to insert subsequent diskettes.
4. The setup program will prompt you to terminate any application that might lock system resources or files. A good practice would be to stop any other application before running setup.
5. The next step will be to select a target directory for the program. It is recommended that the default location be used if possible.
6. Once the target directory has been selected and approved, the setup program initiates the installation procedure. Once the process is over, the program will inform you if the installation was successful or not.

2.3.5 Boot Loader Programming

2.3.5.1 RS232 connection or USB connection

Depending on the technology used to download the Boot Loader application choose one of next paragraphs so that the download can be done correctly. If there are several DMS to be programmed repeat the corresponding procedure once the device has been well addressed.

2.3.5.1.1 Boot Loader programming if RS232 connection is used

To upload the Boot Loader Application in the DMS, you have to execute the following steps:

1. Turn DCU power OFF.
2. Connect PTA computer and DCU with PTA cable.
3. On the Left DMS, install the jumper called "Boot Strap" mode.
4. On the Right DMS, install the jumper called "Reset".
5. Use PTA Application.
6. Select Tools -> Load Flash -> Open Flash File to open the Boot Loader Application.
7. Select DMS Boot Loader and click OK.
8. Select Tools -> Load Flash -> Send Flash File to start to upload the Boot Loader Application. Follow the uploading prompt of the PTA.

9. After Boot Loader Application process is finished, verify if the uploading successful information is presented. If the uploading is successful, follow the prompt to remove the PTA cable, reset the DCU, remove the “Boot Strap” and the “Reset” jumper. If the uploading is failed, redo steps 6 and 7.
10. The DMS is programmed with the Boot Loader Application version.

2.3.5.1.2 Boot Loader programming if USB connection is used

1. Turn DCU power ON.
2. Connect the PTA computer and the DCU with the PTA cable.
3. Start PTA Application.
4. To unlock administration functions, go to menu Administration -> Other Functions -> Password, write **vsrsm8** and press OK.
5. Select Administration -> Other Functions -> Load Flash -> Open Flash File to open the Boot Loader Application.
6. Select Latest version of DMS Boot Loader and click OK.
7. Select Administration -> Other Functions -> Load Flash -> Send Flash File to start to upload the Boot Loader Application.
8. After Boot Loader Application programming is completed, verify if the uploading successful information is presented. If the upload failed, redo steps 6 and 7.
9. The DCU is programmed with the Boot Loader Application version.

2.3.5.2 DMS Application Programming

To upload the DMS Application in the DCU, you have to execute the following steps:

1. Connect PTA computer and DCU with PTA cable.
2. Turn DCU power ON.
3. Use PTA Application.
4. Select Tools -> DCU Upgrade -> Open DMS Application File.
5. Select DMS Application Version and click OK
6. After the file verification, click OK
7. Select Tools -> DCU Upgrade -> Upgrade.
8. Click YES.
9. When the upload is finished successfully, click OK on the prompt window

2.3.6 DMS Application Corruption

To avoid the DMS Boot Loader automatically jump into the DMS Application when the DCU is rebooted, we have to corrupt the DMS Application and let the DMS stay in the Boot Loader. The following procedure is used to corrupt the DMS Application file. The Boot Loader must be uploaded prior to executing this procedure. This procedure is mandatory for proper verification of the Boot Loader.

1. Connect PTA computer and DCU with PTA cable.
2. Turn DCU power ON.
3. Start PTA Application and click on the Connect icon.
4. Select Tools -> DMS Upgrade -> Open DMS Application File.
5. Select DMS Application, Click Open.
6. After the file verification, click OK.
7. Select Tools -> DMS Upgrade -> Upgrade. Click YES.
8. When the upload indicates around 90%, click on Stop.
9. Click OK.
10. Close PTA Application.

2.3.7 Emulator Setup

Many tests use the emulator to simulate faulty conditions. We use Hitex DProbe-ST10 hardware emulator along with Hitop software to stop the DMS Application while it runs and change the value of specific variables. To start the emulator, execute the following steps.

1. Turn OFF the DCU.
2. Remove the DCU cover.
3. Install the emulator PressOn adaptor, as per its installation manual, on the ST10CTRL + B76 CPU.
4. Plug the DProbe-ST10 emulator on the PressOn adaptor; stabilize it with the “leg”.
5. Turn ON the emulator.
6. Turn ON the DCU.

2.3.8 Other Information about Test

In this test procedure, each test section is independent. Within a section, the initial setup is described in the “Special Procedural Requirements” and all test cases must be executed sequentially.

All LEDs described in this test procedure are on the DCU.

When it is required to uses the PTA Application, it is required to log into the target module, establish a session, and open a new project window in the “File” menu. When you close the PTA Application, it is not required to save the project otherwise mentioned in the procedure.

Do all tests with the power supply set to nominal voltage.
Verify with the Voltmeter the supply voltage.

All tests should be performed on one DCU by default, unless otherwise stated.

3 PROCEDURE

In the following test procedure, software applications used are: PTA Application and PTA Simulator.

The PTA and PTA Simulator share one serial communication port (RS-232) and communication to the DCU by PTA cable. There is only one application, either PTA or PTA Simulator, running and communicating to the DCU at a given time. The PTA Simulator should always be in PTA simulation state. Make sure the right serial port number is selected before to use in the test.

To use PTA Application and PTA Simulator, follow test case instructions.

Test Cases inside a section must be done sequentially.

3.1 Login/Logout

Special Procedural Requirements: *Target Module must be programmed with Boot Loader software only. If it's not the case, execute section 2.3.5 , and 2.3.6 on each target modules prior to start this section.*
Use the PTA Simulator for this section. Make sure that the PTA is not running.
The PTA Simulator should be set as the following before testing:

- *DCU check box is unchecked.*
- *Auto check box is unchecked.*
- *PTA Seq. Auto is checked.*
- *Left DCU is checked into DCU Selection.*

Measurements: *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Valid Login on the PTA (PTA_LoginReq) (DMS receives PTA_LoginReq command and DMS answers with PTA_Login message)	Reset the DCU. Select the PTA_LoginReq command in the TX box under RS-232 interface. Send this command.	PTA Simulator is logged in. (Red square close to RX becomes pink.) In the status screen, The PTA Simulator will receive the command PTA_Login. Verify that the 6th byte of the string received (beginning by RX) is 04 (34 XX XX 00 47 **04** 52 31 36 30 2C 20 44 6F 6F 72 20 4D 6F 6E 69 74 6F 72 69 6E 67 20 61 6E 64 20 43 6F 6E 74 72 6F 6C 20 55 6E 69 74 2C 20 56 61 70 6F 72 XX XX).	
2	Invalid Login on the PTA (PTA_LoginReq)	Logout by sending a PTA_LogoutReq. Select the PTA_LoginReq command in the TX box under RS-232 interface. Erase one character to create an invalid login name. Send this command.	Pink square close to RX becomes red. Red square stays red. In the status screen, The PTA Simulator will receive the command PTA_Login. Verify that the 6th byte of the string received (beginning by RX) is 01 (34 XX XX 00 47 **01** 52 31 36 30 2C 20 44 6F 6F 72 20 4D 6F 6E 69 74 6F 72 69 6E 67 20 61 6E 64 20 43 6F 6E 74 72 6F 6C 20 55 6E 69 74 2C 20 56 61 70 6F 72 XX XX)	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	Valid Logout from PTA (PTA_LogoutReq) (DMS receives PTA_LogoutReq command and DMS answer with PTA_Logout message)	Select PTA_LoginReq command and send it. Select the PTA_LogoutReq command. Send this command.	PTA Simulator is logged in. (Red square close to RX becomes pink.) Pink square close to RX becomes red. In the status screen, the PTA Simulator will receive the command PTA_Logout. Verify that the 6th byte of the string received (beginning by RX) is 02 (07 XX XX 00 49 02 XX XX).	
4	Logout from PTA when not logged (PTA_LogoutReq)	Select the PTA_LogoutReq command. Send this command.	In the status screen, the PTA Simulator will receive the command PTA_Logout. Verify that the 6th byte of the string received (beginning by RX) is 00 (07 XX XX 00 49 00 XX XX).	
5	Auto logout active after 60 seconds	Select PTA_LoginReq command and send it. Make sure “Auto” checkbox is not checked. Wait 60 seconds, select PTA_ComTestReq command and send it.	PTA Simulator is logged in. (Red square close to RX becomes pink.) Verify the PTA Simulator receives the command PTA_Logout. Verify that the 6th byte of the string received is 00 (07 XX XX 00 49 00 XX XX).	
6	Auto logout not active between 0s and 60s	Select PTA_LoginReq command and send it. Wait 55 seconds and select PTA_LogoutReq command and send it.	PTA Simulator is logged in. (Red square close to RX becomes pink.) In the status screen, the PTA Simulator will receive the command PTA_Logout. Verify that the 6th byte of the string received is 02 (07 XX XX 00 49 02 XX XX).	

#	Test Cases	Input Specifications	Output Specifications	P/F
7	Valid Login into DMS Application and DMS Boot Loader.	Close PTA Simulator Software. Start the PTA, go to File -> Properties -> Login Option… and select DMS Application Click OK. Reboot DCU. Wait the automatic Login of the PTA Application, less that 60s. In PTA, go to File -> Properties -> Login Option… and select DCU Boot Loader. Click OK. Reboot DCU. Wait the automatic Login of the PTA Application	Verify in the bottom task bar near the connection icon the status is “Boot Loader” because the DCU Application is corrupted. Verify in the bottom task bar near the connection icon the status is in “Boot Loader”	

3.2 Portable Test Application (PTA) Communication Error

Special Procedural Requirements: *DCU must be programmed with DCU Boot Loader software only. If it's not the case, execute sections 2.3.5, and 2.3.6 for left and right controllers prior to start this section.*
Use the PTA Simulator for this section. Make sure that the PTA is not running.
The PTA Simulator should be set as the following before testing:

- *DCU check box is unchecked.*
- *Auto check box is unchecked.*
- *PTA Seq. Auto is checked.*
- *Left DCU is checked into DCU Selection.*

Measurements: *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	DMS To PTA error (PTA_DMStoPTAError)	Reset the DCU. Start PTA Simulator, select PTA_LoginReq command and send it. In the *PTA Seq. Auto,* remove the check and write 9 as sequence number. Select PTA_TimeReq and Send it.	 The red square at the right of RX becomes pink. The DMS send the PTA_ DMStoPTAError message. Verify the 6th byte is 02 and the 7th byte is 40 (08 09 XX 00 32 02 40 XX XX)	

3.3 Fault LED & Self Test

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader software only.*
Execute the Emulator Setup section 2.3.7

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Static RAM test. Valid nVolatile Memory test. Valid CPU internal RAM test. *DMS Application* with good CRC. DMS Boot Loader with good CRC. No Fault found in Boot Loader	Reset the DCU Wait for 30 seconds	 The Warning LED blinks (ON and OFF) at a regular interval. (250ms ON, 250ms OFF)	
2	Volatile Memory Error (Memory cell, address, and data lines error)	Make sure the emulator is correctly plugged in and powered on. Start Hitop application. Open the DMS Boot Loader project. Go to ram.c file and set a breakpoint (click on the left margin) at the line "MemBackupZone.AddressLine.ucCommand = SRAM_ADD_CMD;" (was line 338 in version 3 of the file). Start the emulator by pressing F5 and wait for the program to stop at breakpoint. When the emulator stops at breakpoint, using "go to cursor" jump to line "pucMemoryPosition = (unsigned char *)ulMemoryBase;" (was line 367 in version 3 of the file). When emulator stops, press F10 twice until the line "*pucMemoryPosition = ucData; " is highlighted. Then select the variable ucData and drag it to the watch window. Change its value to 3 (it was 1). Press enter to validate the new value. Press F5 to run the emulaltor. And observe the Fault LED on the DMS board. Reset the emulator by pressing ALT+R.	The breakpoint is inserted. The buttons change to to indicate emulator running. The emulator runs until line "pucMemoryPosition = (unsigned char *)ulMemoryBase;" and stops. The value changes to 3 and becomes red. The buttons change to to indicate emulator is running. The red Fault LED blinks three times to indicate an SRAM failure.	

#	Test Cases	Input Specifications	Output Specifications	P/F
3	nVolatile Memory Error (Memory cell, address, and data lines error. Data preserved during testing)	Open the DMS Boot Loader project. Go to ram.c file and set a breakpoint (click on the left margin) at the line "MemBackupZone.AddressLine.ucCommand = NVRAM_ADD_CMD;" (is line 343 in version 3 of the file). Start the emulator by pressing F5 and wait for the program to stop at breakpoint. When the emulator stops at breakpoint, using "go to cursor" jump to line "pucMemoryPosition = (unsigned char *)ulMemoryBase;" (was line 367 in version 3 o the file). When emulator stops, press F10 twice until the line "*pucMemoryPosition = ucData; "is highlighted. Then select the variable ucData and drag it to the watch window. Change its value to 3 (it was 1). Press enter to validate the new value. Press F5 to run the emulaltor. And observe the Fault LED on the DCU board. Reset the emulator by pressing ALT+R.	The breakpoint is inserted. The buttons change to to indicate emulator running. The emulator runs until line "pucMemoryPosition = (unsigned char *)ulMemoryBase;" and stops. The value changes to 3 and becomes red. The buttons change to to indicate emulator is running. The red Fault LED blinks five times to indicate an NVRAM failure.	

3.4 Target Module Application upgrade

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader software only. If it's not the case, execute section 2.3.5. and 2.3.6 for DMS controllers prior to start this section. Start the PTA, go to File -> Properties -> Login Option… and select DCU Application. Click OK.*
Reboot DCU; the PTA will connect to DMS in Boot Loader mode.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Valid DMS Application upgrade Some functionalities of the communication protocol are tested in this step such as: "Next package required, next package, package transfer and end of packages" PTA_NextPackReq PTA_NextPack PTA_LastPack PTA_FileXferOver	*Use PTA:* Click on Tools, select DCU Upgrade and select Open DCU Application files. Select DMS Application. Click Open. Click OK. Click on Tools menu again and select DCU Upgrade-> Upgrade. Click on YES to confirm DCU upgrade. Verify the upgrade successful window. Click OK to close the window. The DMS Boot Loader will disconnect and start the DMS Application. When the PTA automatically reconnects, look at the DMS App. Version.	 The PTA starts the transfer. Success window is displayed. The PTA disconnects. The Boot Loader transfers te control to the target Application and the PTA will automatically reconnect in DMS Application Mode. The DMS Application is displayed correctly.	

#	Test Cases	Input Specifications	Output Specifications	P/F
2	Invalid DMS Application upgrade	Execute section 2.3.6 to program an incomplete DMS Application. Reboot DMS and wait automatic Login. Click on Connection icon on the bottom of the PTA screen to force DMS Application to be started (Logoff from DMS Boot Loader). Click on Connection icon to establish a session. In the PTA Application, select File -> Properties. In the Login Option box, select Application. Click OK Execute section *2.3.5.2* to program the DMS Application. Reboot the DMS and wait automatic Login.	 Verify the PTA is logged into the DMS Boot Loader. The Warning LED is still flashing (250ms ON, 250ms OFF) because the DMS is still in Boot Loader mode. Verify the PTA displays logging state into DMS Boot Loader. Verify the PTA displays logging state into DCU Application.	
3	PTA with Boot Loader setting selected	In the PTA Application, select File -> Properties. In the Login Option box, select Boot Loader. Click OK Reset the DCU. Wait 30 seconds. Start a stopwatch and stop it after 20 blinks of the Warning LED.	 The PTA is connected in Boot Loader mode. The Warning LED should toggle at 250ms interval. The time is about 10 seconds: 9.5s < ____ < 10.5s	

#	Test Cases	Input Specifications	Output Specifications	P/F
4	PTA with DMS Application setting selected	In the PTA Application, select File -> Properties. In the Login Option box, select DMS Application. Click OK. Reset the DCU. Wait 30 seconds. Close PTA and do not save the project.	The PTA is connected in DMS Application mode. The Normal LED blinks. The Warning LED is either ON or OFF but it does not blink.	

3.5 Software Version

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader and DMS Application software. To do so, execute section 2.3.5, if required prior to start this section.*
Use PTA as requested. Make sure that the PTA simulator is not running when PTA is requested.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	PTA Version Request: PTA Sends PTA_VersionReq Command. DMS answer PTA_SCIVersion.	Reset DCU. *Open PTA Simulator*: Select PTA_LoginReq command and send it. Select PTA_VersionReq with parameter 02 00 (Request all available SCI versions). Send this command *Close PTA Simulator*: *Use PTA:* Click on Login button on the toolbar and the project window will display all software versions and release dates. Close PTA and do not save the project.	The red square at the right of RX becomes pink. The DCU returns PTA_Version with the following content: 2E XX XX 11 02 01 41 50 50 00 00 00 00 00 **XX XX YY ZZ** 31 2E 30 00 00 00 01 01 42 4F 4F 54 00 00 00 00 **XX XX YY ZZ** 31 2E 30 00 00 00 XX XX Compare revisions and versions with the ones used in this test procedure. Boot Version _____ Date_____ Application _____ Date_____	

3.6 Static RAM Memory Dump

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader and DMS Application software. To do so, execute section 2.3.5 if required prior to start this section.*
Start the PTA, go to File -> Properties -> Login Option… and select Boot Loader. Click OK.
Reset the DCU.
The PTA will automatically connect in the Boot Loader mode.
Make sure that the PTA simulator is not running.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Volatile Memory dump Some functionalities of the communication protocol are tested in this step such as: "Next package required, next package, package transfer and end of packages" PTA_NextPackReq PTA_FileXferOver PTA_FileReq PTA_FileStat	*Use PTA:* Click Memory button in the PTA Project window. Click Dump DCU Memory on the Diagnostic menu. Write 1000 in the offset text box. Move the display scroll bar. Close PTA and do not save the project.	The DCU Memory Dump window pops up. After the file transfer is completed, the DCU Memory values are displayed in the DCU Memory Dump window. The display format is: Address—16 bytes data in Hex—16 bytes data in ASCII. The display window will automatically jump to address 0x101000. The display window will automatically jump to related offset address.	

3.7 Request DMS Code Status

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader and DMS Application software. To do so, execute sections 2.3.5 if required prior to start this section.*
Use PTA as requested.
Make sure that the PTA simulator is not running.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Code status request PTA sends to DMS Boot Loader PTA_DMSCodeStatusReq and DMS Boot Loader answers PTA_DMSCodeStatus	*Use PTA:* Establish a session with DMS by clicking the Login button. From the PTA, click on Tools and select DCU Code Status. Verify that the size and the CRC are the same as the ones in the comments of the Application and Boot Loader zip packages. Close PTA and do not save the project.	The DMS returns the name, the size and the CRC of the Boot and the DMS Application software. **ID Name Size CRC** 1 DMS Boot Loader _____ _____ 2 DMS Application _____ _____	

3.8 Door Control Access

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader and DMS Application software. To do so, execute section 2.3.5 if required prior to start this section.*
The PTA Simulator must be closed.
Open the PTA Application.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Door control access by DMS Boot Loader	Make sure the door is closed and locked. In PTA, Select File -> Properties. In the Login Option box, select Boot Loader. Click OK. Reset the DMS. On the DMS test fixture, activate Open/Unlock door signals. In PTA, select File -> Properties. In the Login Option box, select DCU Application. Click OK. Reset the DCU. Wait 15 seconds. On the DCU test fixture, activate Open/Unlock door signals. Close the PTA Application.	The Warning LED blinks (ON and OFF) at a regular interval. The DCU will automatically login in Boot Loader mode. The door does not move. The DMS will automatically login in DMS Application mode. The door opens.	

3.9 DMS Boot Loader upgrade

Special Procedural Requirements: None.
Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Valid DMS Boot Loader upgrade	Upload a Boot Loader application (done in section 2.3.5.1.1).	Done in section 2.3.5.1.1.	
2	Invalid DMS Boot Loader upgrade	Start a Boot Loader application upgrade as per section 2.3.5.1.1. After about 10 seconds of upload, disconnect the PTA cable. Reboot the DCU. Redo the Boot Loader application upgrade as per section 2.3.5.1.1 completely.	The communication is lost and the Boot Loader does not start (the Warning LED does not flash). The DMS is now programmed with Boot Loader application under test.	

3.10 Verification of the Connection State

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader and DMS Application software. To do so, execute sections 2.3.5 if required prior to start this section.*
Use PTA Simulator as requested.
Make sure that the PTA is not running.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Communication Test Message Keep alive signal sent from the PTA to DMS. DMS receives PTA_ComTestReq to Keep alive the Communication. DMS answer with PTA_ComTestStatus message	In PTA Simulator, make sure the Auto checkbox is not checked. Reset DCUSelect PTA_LoginReq command and send it. Wait 55 seconds. Select PTA_ComTestReq command and send it. Wait 65 seconds, select PTA_ComTestReq command and send it.	PTA Simulator is logged in. (Red square close to RX becomes pink.) PTA Simulator receives the command PTA_ComTestStatus (06 XX XX 00 50 XX XX). Verify the PTA Simulator receives the command PTA_Logout. Verify that the 6th byte of the string received is 00 (07 XX XX 00 49 00 XX XX).	

3.11 Controller ID

Special Procedural Requirements: *DMS must be programmed with DMS Boot Loader and DMS Application software. To do so, execute section 2.3.5 if required prior to start this section.*
Use PTA simulator or PTA as requested.

Measurements: *Self-explanatory*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	Controller ID Verification. Test the Controller ID bit. Logging into the Left DMS Controller and then	Use PTA. Make sure DMS Left and DMS Right are running and programmed with DMS Boot Loader and Application by looking at the versions in Project window. Close PTA, don't save the project.	Both Boot Loader and Application versions are the versions under test.	

#	Test Cases	Input Specifications	Output Specifications	P/F
	Verify the Right DMS Controller does not respond to communications	Use PTA Simulator. Uncheck the DCU checkbox. Verify Left DCU is selected in DCU Selection box. Select the PTA_LoginReq command in the TX box under RS-232 interface. Reset the DMS. Send the Login command. Select PTA_ComTestReq command and send it. Change to Right DMS into DCU Selection of PTA Simulator. Select PTA_ComTestReq command and send it. (Following test must be performed within 60 seconds of initial login command above.) Change to Left DMS into DCU Selection of PTA Simulator. Select PTA_ComTestReq command and send it.	PTA Simulator is logged in. (Red square close to RX becomes pink.). PTA Simulator receives the command PTA_ComTestStatus (06 XX XX 00 50 XX XX). Verify the 4th byte of the string is 00 (00 stands for left DMS Boot Loader) Verify the PTA Simulator receives the command PTA_Logout. Verify the 4th byte of the string is 01 and the 6th byte of the string received is 00 (07 XX XX 01 49 00 XX XX). (4th byte, 01 stands for right DMS Boot Loader) PTA_ComTestStatus (06 XX XX 00 50 XX XX). Verify the 4th byte of the string is 00 (00 stands for left DMS Boot Loader)	

3.12 Static Numerical Requirements

Special Procedural Requirements: *DMS must be programmed with DMS application and DMS Boot Loader under test and powered ON.*
Close PTA application when using PTA Simulator.

Measurements: *Self-explanatory.*

#	Test Cases	Input Specifications	Output Specifications	P/F
1	PTA Communication with DMS on RS232 port is 57600 bps, no parity, 8 data bits, and 1 stop bit.	Reset the DCU. *Uses PTA simulator.* Select the PTA_LoginReq command in the TX box under RS-232 interface. Send this command. Verify on the bottom of the PTA simulator window the configuration.	PTA Simulator is logged in. (Red square close to RX becomes pink.) The configuration is COMx, 57600.	
2	DMS Boot Loader uses less than 70% of installed memory (Volatile Memory, nVolatile Memory, separately) check SACS	Calculate the amount of memory marked as “free” in Table 3-1.	Amount of free Volatile Memory: _____>= 38.4 KB Amount of free nVolatile Memory: _____>= 163.2 KB	
3	DMS software (Application and Boot loader) uses less than 70% of installed Flash memory	*Use PTA:* Establish a session with DCU by clicking the Login button. From the PTA, click on Tools and select DCU Code Status. Verify the total size of the DMS Boot Loader and DMS Application is less than 70% of the total installed flash memory (70% of 512Kbytes) => 367000 bytes. Close PTA and do not save the project.	The DMS returns the size and the CRC of the Boot and the DMS Application software. **ID Name Size CRC** #1 DMS Boot Loader _____ _____ #2 DMS Application _____ _____ Total used size (#1+#2): _____	
4	DMS Boot Loader uses less than 50% of the available processing time.	*Use PTA:* Got to menu File->Properties, in Login Option, select Boot Loader. Reset DCU. Click on Diagnostic and select Dump DCU Memory. In the offset box, write 220. This address contains the maximum CPU usage in percent.	Value read at 0x100220: _____ Verify that the value at the address 0x100220 is not greater than 0x32, representing 50%.	

Table 3-1: Memory Mapping Example

Memory type	Start	Stop	Length	Class	Module	Note
Volatile SRAM (SRAM)	0F0000H	0F01FFH	000200H	FCODE	INTERRUPTBOOT	
	0F0220H	0F0220H	000001H	ND_CPU	OS_CONFIG	
	0F0230H	0F1129H	000EFAH	NDATA0	OS_CONFIG	
	0F112AH	0F11AFH	000086H	NDATA0	OS_RESOURCE	
	0F11B0H	0F11C6H	000017H	NDATA0	PTUBOOT	
	0F11C7H	0F11C7H	000001H	NDATA0	BOOT	
	0F11C8H	0F11D7H	000010H	NDATA0	CRC	
	0F11D8H	0F11E3H	00000CH	HDATA0	SERIAL	
	0F1200H	0F12FFH	000100H	N_CSTK	USERSTACK	
	0F1300H	0F1802H	000503H	NRXBUF	SERIAL	
	0F1804H	0F24D9H	000CD6H	HDATA0	PTUBOOT	
	0F24DAH	0F2559H	000080H	HDATA0	0S_RESOURCE	
			00250EH	Total		
			001DAF2	**Free**	**None**	**= 121586 Bytes**
	300000H	3FFFFFH	**100000H**	**Free**	**None**	**= 1048576 Bytes**
Permanent SRAM (NVRAM and Data Flash)	120000H	127EFFH	007F00H	RESERVED (Free)	MEMORY	
	127F00H	127F1DH	00001EH	FD_MEM		
			00001EH	Total		
			007F00H	**Free**	**None**	**= 32512 Bytes**
	400000H	5FFFFFH	**200000H**	**Free**	**None**	**= 2097152 Bytes**

This page is intentionally left blank.

APPENDIX A: TEST REPORT SHEET

(Page 1 of 3)

Item Identification

Item Tested:	
Item P/N:	
Item Version:	

Tester Identification

Tester:	
Time & Date Started:	
Time & Date Finished:	
Physical Location of Test Execution:	

Equipment Used

Description	Part Number	Rev	Serial Number	Calibration Due Date
DMS Test Fixture				
PTA cable DB9-PTA PLUG				
Variable Power Supply (Voltage source min. 10A, 0-50V)				
Multimeter				
PC Running Windows 2000\XP				
Door Monitoring & Control Unit (DMS under test)				
Complete Door System: 1 Door Monitoring and Control Unit (DCU) 2 Door Operators 2 Solenoid Locks 2 Motors				

Software Description	P/N	Version	Date
PTA Application software			
PTA Simulator Software			
DMS Boot Loader software			
DMS Application software			

Test Report Sheet (Page 2 of 3)

Comments

Approvals

Tester: _____ Title: _____ Date: _____

Witness: _____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____
_____ Title: _____ Date: _____

Make Copies of this page and the previous one if required.

Test Report Sheet (Page 3 of 3)

Summary of Anomalies

SPR #	Description